

Actual Salinometer Percent: 3.5%

- 1 point if between 0-10%
- 3 points if between 1-6%
- 3 points if between 2-5%
- 2 points if between 3-4%
- 1 point if 3.5% ←add up for total point amount

MCQ (1 point each):

1. A **B** C D
2. A **B** C D
3. A B **C** D
4. A **B** C D
5. A B **C** D
6. A **B** C D
7. **A** B C D
8. A **B** C D
9. **A** B C D
10. A **B** C D
11. A **B** C D
12. A **B** C D
13. A **B** C D
14. A B C **D**
15. **A** B C D
16. A **B** C D
17. A **B** C D
18. **A** B C D
19. A **B** C D
20. A **B** C D
21. A **B** C D
22. A **B** C D
23. A **B** C D
24. **A** B C D

25. **A** B C D
26. A B **C** D
27. A **B** C D
28. A **B** C D
29. **A** B C D
30. A B **C** D

Short Answer (3 points each):

- **1 pt for identifying correct class or tolerance level**
- **1 pt for explaining what that means for water quality**
- **1 pt for reasoning (oxygen/pollution connection)**

1. Mayflies, caddisflies, and stoneflies
 - a. Class 1 – Pollution Sensitive
 - b. Their presence means clean, well-oxygenated water. They die quickly if pollution or low oxygen occurs, so they indicate excellent water quality.

2. Leeches, midges, and flatworms
 - a. Class 3 – Moderately Tolerant
 - b. Their dominance suggests fair to poor water quality. They can survive low oxygen or moderate pollution, showing that conditions are not ideal for sensitive species.

3. Diversity as an indicator
 - a. A stream with many species from Classes 1–2 has good water quality.
 - b. If mostly Class 4–5 species, the water is polluted or low in oxygen.
 - c. High diversity across sensitivity levels = healthier ecosystem.

4. Aquatic sowbugs and scuds
 - a. Class 2 – Moderately Sensitive
 - b. Indicate fair/good quality water with some organic matter and moderate oxygen. They can tolerate slight pollution but not heavy contamination.

5. Air-breathing insects (water striders, whirligig beetles)
 - a. Class 5 – Air Breathing / Pollution Tolerant
 - b. They get oxygen directly from the air, so they can live in polluted or low-oxygen water, unlike mayflies that depend on dissolved oxygen.
6. Changes over time
 - a. If pollution increases, Class 4–5 organisms (leeches, midges, air-breathers) increase and sensitive ones disappear.
 - b. If pollution decreases and oxygen improves, Class 1–2 species return, showing recovery of water quality.

True or False:

1. **False**
2. **True**
3. **False**
4. **True**
5. **True**
6. **False**
7. **False**
8. **True**
9. **True**
10. **False**
11. **False**
12. **True**
13. **True**
14. **True**
15. **True**

Nitrogen Cycle Diagram:

1. Assimilation
2. Nitrates (NO_3^-)
3. Decomposers
4. Ammonium (NH_4^+)

Table: Water sample results from Oak Creek.

1. Site 3 (downstream)
2. Site 3 (downstream)
3. As nitrates and phosphates increase, dissolved oxygen decreases.
4. Site 3 (downstream)
5. Possible answers: Agricultural runoff (fertilizers containing nitrate/phosphate), Urban stormwater runoff (oil, fertilizers, detergents), Wastewater discharge (septic or treatment plant effluent)

Speed ID:

1. Mayfly - Pollution Sensitive (Class 1)
2. Dobsonfly - Pollution Sensitive (Class 1)
3. Purple Loosestrife - Aquatic Nuisance Plant
4. Scud - Moderately Sensitive (Class 2)
5. Giant Water Bug - Air Breathing (Class 5)
6. Zebra Mussel - Aquatic Nuisance Animal
7. Midge - Moderately Tolerant (Class 3)
8. Gilled Snail - Pollution Sensitive (Class 1)

9. Predaceous Diving Beetle - Air Breathing (Class 5)
10. Whirligig Beetle - Air Breathing (Class 5)
11. Tubifex - Pollution Tolerant (Class 4)
12. Eurasian Water Milfoil - Aquatic Nuisance Plant
13. Damselfly - Moderately Sensitive (Class 2)
14. Flatworm - Moderately Tolerant (Class 3)
15. Asian Carp - Aquatic Nuisance Animal